6.1

a)

$P(X=x_1) = 0.01 + 0.05 + 0.1 = 0.16$

$P(X=x_2) = 0.02 + 0.1 + 0.05 = 0.17$

$P(X=x_3) = 0.03 + 0.05 + 0.03 = 0.11$

$P(X=x_4) = 0.1 + 0.07 + 0.05 = 0.22$

$P(X=x_5) = 0.1 + 0.2 + 0.04 = 0.34$

b)

$P(Y=y_1) = 0.01 + 0.02 + 0.03 + 0.1 + 0.1 = 0.26 = \sum_{i=1}^{5} P(Y=y_1, X=x_i)$

$P(Y=y_2) = 0.05 + 0.1 + 0.05 + 0.07 + 0.2 = 0.47$

$P(Y=y_3) = 0.1 + 0.05 + 0.03 + 0.05 + 0.04 = 0.27$

b)

$$P(X=x_1 \mid Y=y_1) = \frac{P(X=x_1, Y=y_1)}{P(Y=y_1)} = \frac{0.01}{0.26} \approx 0.038$$

$$P(X=x_2 \mid Y=y_1) = \frac{P(X=x_2, Y=y_1)}{P(Y=y_1)} = \frac{0.02}{0.26} \approx 0.077$$

$$P(X=x_3 \mid Y=y_1) = \frac{P(X=x_3, Y=y_1)}{P(Y=y_1)} = \frac{0.03}{0.26} \approx 0.115$$

$$P(X=x_4 \mid Y=y_1) = \frac{P(X=x_4, Y=y_1)}{P(Y=y_1)} = \frac{0.1}{0.26} \approx 0.38$$

$$P(x=x_5 \mid y=y_1) = \frac{P(x=x_5, y=y_1)}{P(y=y_1)} = \frac{0.1}{0.26} \approx 0.38$$

$$P(y=y_1 \mid x=x_3) = \frac{P(y=y_1, x=x_3)}{P(x=x_3)} = \frac{0.03}{0.11} \approx 0.27$$

$$P(y=y_2 \mid x=x_3) = \frac{P(y=y_2, x=x_3)}{P(x=x_3)} = \frac{0.07}{0.11} \approx 0.64$$

$$P(y=y_3 \mid x=x_3) = \frac{P(y=y_3, x=x_3)}{P(x=x_3)} = \frac{0.03}{0.11} \approx 0.27$$

6.2

a) $p(x_1) = \int p(x_1, x_2)\, dx_2$

$$p(x_1) = \int 0.4\, \mathcal{N}\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \middle| \begin{bmatrix} 10 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right) dx_2 + \int 0.6\, \mathcal{N}\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \middle| \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 8.4 & 2.0 \\ 2.0 & 1.7 \end{bmatrix}\right) dx_2$$

$$= 0.4\, \mathcal{N}(10, 1) + 0.6\, \mathcal{N}(0, 8.4)$$

$$p(x_2) = \int 0.4\, \mathcal{N}\left(\begin{bmatrix} 10 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right) dx_1 + \int 0.6\, \mathcal{N}\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 8.4 & 2.0 \\ 2.0 & 1.7 \end{bmatrix}\right) dx_1$$

$$p(x_2) = 0.4\, \mathcal{N}(x_2 \mid 2, 1) + 0.6\, \mathcal{N}(x_2 \mid 0, 1.7)$$

b)

$$E_{x_1}[x_1] = \int x_1\, p(x_1)\, dx_1$$

$$E_{x_1}[x_1] = \int x_1 \left[0.4\, N(x_1|10,1) + 0.6\, N(x_1|0, 8.4)\right] dx_1$$

$$= 0.4 \underbrace{\int x_1 N(x_1|10,1)\, dx_1}_{\text{expected value of 1st Gaussian}} + 0.6 \underbrace{\int x_1 N(x_1|0,8.4)\, dx_1}_{\text{expected value of 2nd Gaussian}}$$

$$= 0.4 \cdot 10 + 0.6 \cdot 0 = 4$$

median $p(x_1)$:

$$\int_{-\infty}^{t} p(x_1)\, dx_1 = \int_{t}^{\infty} p(x_1)\, dx_1 = 0.5$$

No analytical solution, numerical solution ≈ 2.8

median $p(x_2)$:

$$\int_{-\infty}^{t} p(x_2)\, dx_2 = \int_{t}^{\infty} p(x_2)\, dx_2 = 0.5$$

again, no analytical solution, numerical solution ≈ 0.87

mode $p(x_1)$:

We need to find where $\frac{dp(x_1)}{dx_1} = 0$ and $\frac{d^2p(x_1)}{dx_1^2} < 0$, so it is a local maximum

numerically: 0.0, 9.998 two modes

mode $p(x_2)$:

$$\frac{dp(x_2)}{dx_2} = 0 \quad \text{and} \quad \frac{d^2p(x_2)}{dx_2^2} < 0$$

numerically: 1.33

c)

The mean for the 2D Gaussian is

$$0.4\begin{bmatrix}10\\2\end{bmatrix} + 0.6\begin{bmatrix}0\\0\end{bmatrix} = \begin{bmatrix}4\\0.8\end{bmatrix}$$

The modes for the 2D Gaussian are local maxima where $\nabla f = 0$ and $|H| > 0$ and $f_{xx} < 0$

where $f_{xx} = \frac{\partial^2 f}{\partial x^2} = H_{11}$

Numerically: $\begin{bmatrix}0.01\\0.01\end{bmatrix}, \begin{bmatrix}9.990\\2.014\end{bmatrix}$

6.3

$$p(\mu \mid x_1, \dots, x_N) = \frac{p(x_1, \dots, x_n \mid \mu)\, p(\mu)}{p(x_1, \dots, x_N)}$$

$$p(x_1, \dots x_n \mid \mu) = \prod_{i=1}^{N} p(x_i \mid \mu) = \prod_{i=1}^{N} \mu^x (1-\mu)^{1-x} = \mu^{ix} (1-\mu)^{i(1-x)}$$

The Beta distribution is a conjugate prior, so $p(\mu) \propto \mu^{\alpha-1} (\mu-1)^{\beta-1}$

The denominator $p(x_1, \dots, x_N) = \int p(x_1, \dots, x_N, \mu)\, d\mu$ which is a normalization factor.

$$p(x_1, \dots x_N \mid \mu)\, p(\mu) = \mu^{ix} (1-\mu)^{i(1-x)} \propto^{\alpha-1} (\mu-1)^{\beta-1} =$$

$$= \mu^{(\alpha + ix)-1} (1-\mu)^{(\beta + i(1-x))-1}$$

Which means $p(\mu \mid x_1, \dots x_N) = \text{Beta}\left(\alpha + ix,\ \beta + i(1-x)\right)$

So the posterior distribution has the same form as the prior.

(6.4) Let C be a random variable that represents the "heads" and "tails" state of the coil.

$$P(C = \text{"heads"}) = \frac{6}{10} \qquad P(C = \text{"tails"}) = \frac{4}{10}$$

Let F be a random variable used to record what fruit we get, either "mango" or "apple".

We can now get conditional distributions for F.

Since when we get C = "heads", we pick from bag 1, we have

$$P(F = \text{mango} \mid C = \text{heads}) = \frac{4}{6} \qquad P(F = \text{apple} \mid C = \text{heads}) = \frac{2}{6}$$

$$P(F = \text{mango} \mid C = \text{tails}) = \frac{4}{8} \qquad P(F = \text{apple} \mid C = \text{tails}) = \frac{4}{8}$$

We got a mango, so we have to find what bag it came from, i.e. we need to find $P(C = \text{tails} \mid F = \text{mango})$

$$P(C = \text{tails} \mid F = \text{mango}) = \frac{P(F = \text{mango} \mid C = \text{tails})\, P(C = \text{tails})}{P(F = \text{mango})}$$

where

$$P(F = \text{mango} \mid C = \text{tails}) = \frac{4}{8} = \frac{1}{2}$$

$$P(C = \text{tails}) = \frac{4}{10}$$

$$P(F = \text{mango}) = P(F = \text{mango}, C = \text{heads}) + P(F = \text{mango}, C = \text{tails}) =$$

$$= \frac{2}{3} \cdot \frac{6}{10} + \frac{1}{2} \cdot \frac{4}{10} = \frac{4}{10} + \frac{2}{10} = \frac{6}{10}$$

$$P(C = \text{tails} \mid F = \text{mango}) = \frac{\frac{1}{2} \cdot \frac{4}{10}}{6/10} = \frac{2}{10} \cdot \frac{10}{6} = \frac{1}{3}$$

6.5

$$x_{t+1} = Ax_t + w \qquad w \sim N(0, Q)$$

$$y_t = x_t + v \qquad v \sim N(0, R)$$

$$p(x_0) = N(\mu_0, \Sigma_0)$$

a) Let's start with $p(x_0, x_1)$ only

$$p(x_0, x_1) = p(x_1, x_0) = p(x_1 | x_0)\, p(x_0)$$ using product rule.

For $p(x_1|x_0)$ we know the value for x_0, so Ax_0 is deterministic.

The only source of randomness is w.

So we have $x_1 = Ax_0 + w$ when Ax_0 is a constant.

Adding a constant Ax_0 to the normally distributed w just shifts the mean, so $Ax_0 + w \sim N(Ax_0, Q)$, so $x_1 \sim N(Ax_0, Q)$.

Since $p(x_1|x_0)$ is Gaussian and x_0 is Gaussian, then

the product $p(x_1|x_0)\,p(x_0)$ is Gaussian, so $p(x_1, x_0)$ is Gaussian. From this it follows that $p(x_0, x_1, \ldots, x_T)$ is a Gaussian distribution.

Next element:

$$p(x_0, x_1, x_2) = p(x_2 | x_0, x_1)\, p(x_0, x_1)$$

$$x_2 = A(\underbrace{Ax_0}_{x_1}) + 2w$$

Again $A(Ax_0)$ is a constant, but in this case the variance has increased.

$$p(x_2 | x_0, x_1) \sim N(AAx_0, 2Q)$$

where the $2Q$ comes from adding $w + w$, since the variances are added when summing two Gaussian distributions

b)

$$p(x_t \mid y_1, \dots y_t) = N(\mu_t, \Sigma_t)$$

①

$$p(x_1 \mid y_1) = N(\mu_1, \Sigma_1)$$

Since $x_2 = Ax_1 + w$, we can now compute $p(x_2 \mid y_1)$

From eq 6.86 and 6.87 we get $Ax_1 \sim N(A\mu_1, A\Sigma_1 A^T)$

and addy the w, we get $x_2 \sim N(A\mu_1, A\Sigma_1 A^T + Q)$

Applyg to x_{t+1} we get

$$x_{t+1} = Ax_t + w \longrightarrow x_{t+1} \sim N(A\mu_t, A\Sigma_t A^T + Q)$$

②

$$p(x_{t+1}, y_{t+1} \mid y_1, \dots, y_t) = p(y_{t+1} \mid x_{t+1}, y_1, \dots, y_t) \cdot \underbrace{p(x_{t+1} \mid y_1, \dots, y_t)}_{\text{computed in part 1}}$$

So we only need to calculate

$$p(y_{t+1} \mid x_{t+1}, y_1, \dots, y_t)$$

we have $y_{t+1} = Cx_{t+1} + v$ and we are given x_{t+1}, so

Cx_{t+1} is a constant, so $y_{t+1} \sim N(Cx_{t+1}, R)$

then $p(x_{t+1}, y_{t+1} \mid y_1, \dots, y_t) = N(Cx_{t+1}, R) \cdot N(A\mu_t, A\Sigma_t A^T + Q)$

Usg Eq. 6.74-6.76 we get

$$C = \left(R^{-1} + (A\Sigma_t A^T + Q)^{-1}\right)^{-1}$$

$$p(x_{t+1}, y_{t+1} \mid y_1, \dots, y_t) \sim c\, N\left(C\left(R^{-1} C x_{t+1} + (A\Sigma_t A^T + Q)^{-1} A\mu_t\right), C\right)$$

where c is given by 6.76.

③

$$p(x_{t+1} \mid y_1, \dots, y_t, y_{t+1}) = \frac{p(x_{t+1}, y_{t+1} \mid y_1, \dots, y_t)}{p(y_{t+1} \mid y_1, \dots, y_t)} \leftarrow \text{part 2}$$

From part 1, we have $p(x_{t+1} \mid y_1, \dots, y_t) = \mathcal{N}(A\mu_t, A\Sigma_t A^T + Q)$

and $y_{t+1} = C x_{t+1} + v$

$$p(y_{t+1} \mid y_1, \dots, y_t) = \mathcal{N}(CA\mu_t, C(A\Sigma_t A^T + Q)C^T + R)$$

Then $$p(x_{t+1} \mid y_1, \dots, y_{t+1}) = \frac{\mathcal{N}(Cx_{t+1}, R) \cdot \mathcal{N}(A\mu_t, A\Sigma_t A + Q)}{\mathcal{N}(CA\mu_t, C(A\Sigma_t A^T + Q)C^T + R)}$$

6.6

Prove $\mathbb{V}_x[x] = \mathbb{E}_x[x^2] - (\mathbb{E}_x[x])^2$

We have $\mathbb{V}_x[x] = \mathbb{E}_x[(x-\mu)^2]$

$\mathbb{E}_x[g(x)] = \int g(x)p(x)\,dx$, so $\mathbb{E}_x[(x-\mu)^2] = \int (x-\mu)^2 p(x)\,dx$

$$\int (x-\mu)^2 p(x)\,dx = \int (x^2 - 2x\mu + \mu^2) p(x)\,dx =$$

$$= \int x^2 p(x)\,dx - \int 2x\mu\, p(x)\,dx + \int \mu^2 p(x)\,dx =$$

$$= \mathbb{E}_x[x^2] - 2\mu \int x p(x)\,dx + \mu^2 \int \underbrace{p(x)\,dx}_{=1}$$

$$= \mathbb{E}_x[x^2] - 2\mu\,\mathbb{E}_x[x] + \mu^2 = \mathbb{E}_x[x^2] - 2\mu^2 + \mu^2 =$$

$\mu = \mathbb{E}_x[x]$ by definition

$$= \mathbb{E}_x[x^2] - \mu^2 = \mathbb{E}_x[x^2] - (\mathbb{E}[x])^2$$

6.7

$$\frac{1}{N^2}\sum_{i,j=1}^{N}(x_i-x_j)^2 = \frac{1}{N^2}\sum_{i,j=1}^{N} x_i^2 - 2x_ix_j + x_j^2 =$$

$$= \frac{1}{N^2}\left[\sum_{i=1}^{N}\sum_{j=1}^{N} x_i^2 - \sum_{i=1}^{N}\sum_{j=1}^{N} 2x_ix_j + \sum_{i=1}^{N}\sum_{j=1}^{N} x_j^2\right]$$

$$\sum_{i=1}^{N}\sum_{j=1}^{N} x_i^2 = \sum_{j=1}^{N}\sum_{i=1}^{N} x_i^2 = N\sum_{i=1}^{N} x_i^2 \quad , \text{ similary } \sum_{i=1}^{N}\sum_{j=1}^{N} x_j^2 = \sum_{j=1}^{N}\sum_{i=1}^{N} x_i^2 = N\sum_{i=1}^{N} x_i^2$$

$$-\sum_{i=1}^{N}\sum_{j=1}^{N} 2x_ix_j = -2\sum_{i=1}^{N}\sum_{j=1}^{N} x_ix_j = -2\sum_{i=1}^{N}\left(x_i\cdot\sum_{j=1}^{N} x_j\right) = -2\left(\sum_{j=1}^{N} x_j\right)\left(\sum_{i=1}^{N} x_i\right) =$$

$$= -2\left(\sum_{i=1}^{N} x_i\right)^2$$

this part does not depend on i, so can move in front

Substituty these in we get:

$$\frac{1}{N^2}\left[2N\sum_{i=1}^{N} x_i^2 - 2\left(\sum_{i=1}^{N} x_i\right)^2\right] = 2\left[\frac{N}{N^2}\sum_{i=1}^{N} x_i^2 - \frac{1}{N^2}\left(\sum_{i=1}^{N} x_i\right)^2\right] =$$

$$= 2\left[\frac{1}{N^2}\sum_{i=1}^{N} x_i^2 - \left(\frac{1}{N}\sum_{i=1}^{N} x_i\right)^2\right]$$

6.8

$$p(x|\mu) = \mu^x(1-\mu)^{1-x}$$

$$= \exp\left(\log\left[\mu^x(1-\mu)^{1-x}\right]\right) =$$

$$= \exp\left(x\log\mu + (1-x)\log(1-\mu)\right)$$

$$= \exp\left(x\log\mu + \log(1-\mu) - x\log(1-\mu)\right)$$

$$= \exp\left(x\log\mu - x\log(1-\mu) + \log(1-\mu)\right) = \exp\left(x\log\left(\frac{\mu}{1-\mu}\right) + \log(1-\mu)\right)$$

$\phi(x) = x$ $\qquad \theta = \log\left(\frac{\mu}{1-\mu}\right)$ $\qquad A(\theta) = -\log(1-\mu)$

$$= \exp\left(x\log\left(\frac{\mu}{1-\mu}\right) + \log(1-\mu)\right)$$

(6.9) Binomial:

$$p(x \mid N, \mu) = \binom{N}{x} \mu^x (1-\mu)^{N-x}$$

$$= \exp\left(\log\left[\binom{N}{x} \mu^x (1-\mu)^{N-x}\right]\right) =$$

$$= \exp\Big(\underbrace{\log\binom{N}{x}}_{\log(h(x))} + x\log\mu + (N-x)\log(1-\mu)\Big)$$

$$= \exp\Big(\log\binom{N}{x} + x\log\mu + N\log(1-\mu) - x\log(1-\mu)\Big)$$

$$A(\theta) = -N\log\left(1 - \frac{1}{1+e^{-\theta}}\right)$$

$$= -N\log\left(\frac{1}{1+e^{\theta}}\right) = N\log(1+e^{\theta})$$

$$A(\theta) = N\log(1+e^{\theta})$$

$$= \exp\Big(\underbrace{\log\binom{N}{x}}_{\log(h(x))} + x\log\left(\frac{\mu}{1-\mu}\right) + N\log(1-\mu)\Big)$$

$$= \binom{N}{x} \exp\left(x\log\frac{\mu}{1-\mu} + N\log(1-\mu)\right)$$

$$\phi(x) = x \qquad \theta = \log\frac{\mu}{1-\mu} \qquad \mu = \frac{e^{\theta}}{1+e^{\theta}} = \frac{1}{1+e^{-\theta}}$$

Beta:

$$p(x \mid \alpha, \beta) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1-x)^{\beta-1} = f(\alpha,\beta)\, x^{\alpha-1}(1-x)^{\beta-1}$$

$$\exp\left(\log\left[f(a,b)\, x^{\alpha-1}(1-x)^{\beta-1}\right]\right)$$

$$= \exp\Big(\log f(a,b) + (\alpha-1)\log x + (\beta-1)\log(1-x)\Big)$$

$$\theta = \begin{bmatrix} \alpha-1 \\ \beta-1 \end{bmatrix} \qquad \phi(x) = \begin{bmatrix} \log x \\ \log(1-x) \end{bmatrix}$$

$$A(\theta) = -\log\left(\frac{\Gamma(\theta_1+1, \theta_2+1)}{\Gamma(\theta_1+1)\Gamma(\theta_2+1)}\right)$$

$$\theta_1 + 1 = \alpha$$

$$\theta_2 + 1 = \beta$$

Binomial likelihood · Beta prior:

$$\binom{N}{x} \exp\left(x\log\frac{\mu}{1-\mu} + N\log(1-\mu)\right) \cdot \exp\Big((\alpha-1)\log x + (\beta-1)\log(1-x) + \log f(\alpha,b)\Big)$$

$$= \binom{N}{x} \exp\left(x\log\frac{\mu}{1-\mu} + (\alpha-1)\log x + (\beta-1)\log(1-x) + \left[N\log(1-\mu) + \log f(\alpha,\beta)\right]\right)$$

$$\theta = \begin{bmatrix} \left(\log\frac{\mu}{1-\mu}\right) \\ (\alpha-1) \\ (\beta-1) \end{bmatrix} \qquad \phi(x) = \begin{bmatrix} x \\ \log x \\ \log(1-x) \end{bmatrix}$$

$$h(x) = \binom{N}{x}$$

$$\theta_1 = \log \frac{\mu}{1-\mu}$$

$$\theta_2 = \alpha - 1$$

$$\theta_3 = \beta - 1$$

$$A(\theta) = -\log f(\alpha, \beta) - N \log(1-\mu)$$

$$= \log \frac{1}{f(\alpha,\beta)} - N \log(1-\mu)$$

$$= \log \frac{1}{f(\theta_2+1, \theta_3+1)} + N \log(1+e^{\theta_1})$$

$$= \frac{\Gamma(\theta_2+1)\Gamma(\theta_3+1)}{\Gamma(\theta_2+1, \theta_3+1)} + N \log(1+e^{\theta_1})$$

6.10

$$p(x \mid a, A) = (2\pi)^{-D/2} |A|^{-\frac{1}{2}} \exp\left[-\frac{1}{2}\left((x-a)^T A^{-1} (x-a)\right)\right]$$

$$p(x \mid b, B) = (2\pi)^{-\frac{D}{2}} |B|^{-\frac{1}{2}} \exp\left[-\frac{1}{2}\left((x-b) B^{-1} (x-b)\right)\right]$$

$$p(x \mid a, A) \cdot p(x \mid b, B) = (2\pi)^{-D} |A|^{-\frac{1}{2}} |B|^{-\frac{1}{2}} \cdot \exp\left[-\frac{1}{2}\left((x-a)^T A^{-1}(x-a) + (x-b)^T B^{-1}(x-b)\right)\right]$$

focusing on the exponent...

$$-\frac{1}{2}\left((x-a)^T A^{-1}(x-a) + (x-b)^T B^{-1}(x-b)\right) =$$

$$= -\frac{1}{2}\left(x^T A^{-1} x - x^T A^{-1} a - a^T A^{-1} x + a^T A^{-1} a + x^T B^{-1} x - x^T B^{-1} b - b^T B^{-1} x + b^T B^{-1} b\right)$$

$$= -\frac{1}{2}\left[x^T (A^{-1} + B^{-1}) x - x^T (A^{-1} a + B^{-1} b) - (a^T A^{-1} + b^T B^{-1}) x + a^T A^{-1} a + b^T B^{-1} b\right]$$

Using 6.74 - $C = (A^{-1} + B^{-1})^{-1} \rightarrow C^{-1} = (A^{-1} + B^{-1})$

6.75 $c = C(A^{-1} a + B^{-1} b) \rightarrow c^T = (a^T A^{-1} + b^T B^{-1}) C$

note that A, B, C and their inverses are square symmetric matrices. so $A^T = A$, $(A^{-1})^T = A^{-1}$

$$C^{-1} c = C^{-1} C (A^{-1} a + B^{-1} b) = A^{-1} a + B^{-1} b$$

$$c^T C^{-1} = (a^T A^{-1} + b^T B^{-1}) C C^{-1} = a^T A^{-1} + b^T B^{-1}$$

$$= -\frac{1}{2}\left[x^T C^{-1} x - x^T C^{-1} c - c^T C^{-1} x + a^T A^{-1} a + b^T B^{-1} b\right]$$

$$= -\frac{1}{2}\left[x^T C^{-1} x - x^T C^{-1} c - c^T C^{-1} x + a^T A^{-1} a + b^T B^{-1} b + c^T C^{-1} c - c^T C^{-1} c\right]$$

$$= -\frac{1}{2}\left[x^T C^{-1} x - x^T C^{-1} c - c^T C^{-1} x + c^T C^{-1} c + a^T A^{-1} a + b^T B^{-1} b - c^T C^{-1} c\right]$$

$$= -\frac{1}{2}\left[(x-c)^T C^{-1} (x-c) + a^T A^{-1} a + b^T B^{-1} b - c^T C^{-1} c\right]$$

Focusing on $a^T A^{-1} a + b^T B^{-1} b - c^T C^{-1} c$

$$a^T A^{-1} a + b^T B^{-1} b - (a^T A^{-1} + b^T B^{-1}) \cancel{C} \cancel{C^{-1}} C (A^{-1} a + B^{-1} b) =$$

$$= a^T A^{-1} a + b^T B^{-1} b - (a^T A^{-1} + b^T B^{-1}) C (A^{-1} a + B^{-1} b)$$

$$= a^T A^{-1} a + b^T B^{-1} b - a^T A^{-1} C A^{-1} a - a^T A^{-1} C B^{-1} b - b^T B^{-1} C A^{-1} a - b^T B^{-1} C B^{-1} b$$

Focusing on $a^T A^{-1} a - a^T A^{-1} C A^{-1} a =$

$$= a^T (A^{-1} - A^{-1} C A^{-1}) a = a^T \left[A^{-1}(I - C A^{-1})\right] a$$

$$= a^T \left[A^{-1}\left(I - (A^{-1} + B^{-1})^{-1} A^{-1}\right)\right] a \qquad (A^{-1} + B^{-1})^{-1} = \left(A^{-1}(A+B)B^{-1}\right)^{-1} = B(A+B)^{-1} A$$

$$= a^T \left[A^{-1}\left(I - B(A+B)^{-1} A A^{-1}\right)\right] a$$

$$= a^T \left[A^{-1}\left(I - B(A+B)^{-1}\right)\right] a$$

$$= a^T \left[A^{-1}\left((A+B)(A+B)^{-1} - B(A+B)^{-1}\right)\right] a$$

$$= a^T \left[A^{-1}\left((A+B-B)(A+B)^{-1}\right)\right] a = a^T \left[\cancel{A^{-1}}(\cancel{A})(A+B)^{-1}\right] a = a^T (A+B)^{-1} a$$

The same derivation can be used for $b^T B^{-1} b - b^T B^{-1} C B^{-1} b$

where we get $b^T (A+B)^{-1} b$

The remaining two terms are $-a^T A^{-1} C B^{-1} b - b^T B^{-1} C A^{-1} a$

$$-a^T A^{-1} (A^{-1}+B^{-1})^{-1} B^{-1} b \qquad (A^{-1}+B^{-1})^{-1} = \left(B^{-1}(A+B)A^{-1}\right)^{-1}$$

$$= A(A+B)^{-1} B$$

$$= B(A+B)^{-1} A$$

$$-a^T \cancel{A^{-1}} \cancel{A} (A+B)^{-1} \cancel{B} \cancel{B^{-1}} b = -a^T (A+B)^{-1} b$$

$$-b^T B^{-1} (A^{-1}+B^{-1})^{-1} A^{-1} a = -b^T \cancel{B^{-1}} \cancel{B} (A+B)^{-1} \cancel{A} \cancel{A^{-1}} a = -b^T (A+B)^{-1} a$$

So we have:

$$a^T (A+B)^{-1} a + b^T (A+B)^{-1} b - a^T (A+B)^{-1} b - b^T (A+B)^{-1} a =$$

$$= (a-b)^T (A+B)^{-1} (a-b)$$

Going back to $p(x|a,A) \cdot p(x|b,B)$

$$= (2\pi)^{-D} |A|^{-\frac{1}{2}} |B|^{-\frac{1}{2}} \exp\left(-\frac{1}{2}\left[(x-c)^T C^{-1} (x-c) + (a-b)^T (A+B)^{-1} (a-b)\right]\right)$$

$$= \underbrace{(2\pi)^{-D} |A|^{-\frac{1}{2}} |B|^{-\frac{1}{2}} \exp\left(-\frac{1}{2}\left[(x-c)^T C^{-1} (x-c)\right]\right)}_{\mathcal{N}(x|c,C)} \cdot \exp\left(-\frac{1}{2}\left[(a-b)^T (A+B)^{-1} (a-b)\right]\right)$$

$$= \underbrace{(2\pi)^{-\frac{D}{2}} |C|^{-\frac{1}{2}} \exp\left(-\frac{1}{2}\left[(x-c)^T C^{-1} (x-c)\right]\right)}_{c} \cdot \underbrace{(2\pi)^{-\frac{D}{2}} \cdot \frac{|AB|^{-\frac{1}{2}}}{|C|^{-\frac{1}{2}}} \cdot \exp\left(-\frac{1}{2}\left[(a-b)^T (A+B)^{-1} (a-b)\right]\right)}_{c}$$

$$\frac{|AB|^{-\frac{1}{2}}}{|C|^{-\frac{1}{2}}} = \left(\frac{|AB|}{|C|}\right)^{-\frac{1}{2}} = \left(|AC^{-1}B|\right)^{-\frac{1}{2}} = \left(|A(A^{-1}+B^{-1})B|\right)^{-\frac{1}{2}} = |A+B|^{-\frac{1}{2}}$$

$$C^{-1} = (A^{-1}+B^{-1})$$

which gives us

$$(2\pi)^{-\frac{D}{2}} |A+B|^{-\frac{1}{2}} \exp\left(-\frac{1}{2}(a-b)^T (A+B)^{-1} (a-b)\right) = c$$

6.10

b)

$$p(x/\mu, \Sigma) = (2\pi)^{-\frac{D}{2}} |\Sigma|^{-\frac{1}{2}} \exp\left(-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right)$$

We need to express this in the natural parameters form

$$\exp\left(\log\left((2\pi)^{-\frac{D}{2}} |\Sigma|^{-\frac{1}{2}}\right)\right) \exp\left(-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right)$$

$$= \exp\left(-\frac{D}{2}\log(2\pi) - \frac{1}{2}\log(|\Sigma|)\right)\exp\left(-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right)$$

$$= \exp\left[-\frac{1}{2}x^T\Sigma^{-1}x + \frac{1}{2}x^T\Sigma^{-1}\mu + \frac{1}{2}\mu^T\Sigma^{-1}x - \frac{1}{2}\mu^T\Sigma^{-1}\mu - \frac{D}{2}\log(2\pi) - \frac{1}{2}\log(|\Sigma|)\right]$$

$$= \exp\left[-\frac{1}{2}x^T\Sigma^{-1}x + \mu^T\Sigma^{-1}x - \frac{1}{2}\left(D\log(2\pi) + \log(|\Sigma|) + \mu^T\Sigma^{-1}\mu\right)\right]$$

when $\mu^T\Sigma^{-1}x = x^T\Sigma^{-1}\mu$ since $(\mu^T\Sigma^{-1}x)^T = x^T\Sigma^{-1}\mu$ and this evaluates to a scalar so they must be equal.

$$= \exp\left[tr\left(-\frac{1}{2}\Sigma^{-1}xx^T\right) + \mu^T\Sigma^{-1}x - \frac{1}{2}\left(D\log(2\pi) + \log(|\Sigma|) + \mu^T\Sigma^{-1}\mu\right)\right]$$

when $-\frac{1}{2}x^T\Sigma^{-1}x = -\frac{1}{2}tr(\Sigma^{-1}xx^T) = tr\left(-\frac{1}{2}\Sigma^{-1}xx^T\right) = vec\left(-\frac{1}{2}\Sigma^{-1}\right)^T vec(xx^T)$

$$vec\left(-\frac{1}{2}\Sigma^{-1}\right) = \begin{bmatrix} \Sigma_{11} \\ \vdots \\ \Sigma_{1d} \\ \Sigma_{21} \\ \vdots \\ \Sigma_{dd} \end{bmatrix} \quad \text{and} \quad vec(xx^T) = \begin{bmatrix} x_1^2 \\ \vdots \\ x_1 \times d \\ x_2 x_1 \\ \vdots \\ x_d^2 \end{bmatrix}$$

We can express

$$\mathrm{vec}\left(-\tfrac{1}{2}\Sigma^{-1}\right)\cdot \mathrm{vec}\left(xx^T\right) + \mu^T\Sigma^{-1}x$$ as a single dot product

by concatenating into a large vector

$$\begin{bmatrix} \Sigma^{-1}\mu \\ \mathrm{vec}\left(-\tfrac{1}{2}\Sigma^{-1}\right) \end{bmatrix} \cdot \begin{bmatrix} x \\ \mathrm{vec}\left(xx^T\right) \end{bmatrix} = \langle \theta, \phi(x) \rangle$$

So now if we have $\mathcal{N}(x \mid a, A)\cdot \mathcal{N}(x \mid b, B)$ we get

$$\exp\left(\begin{bmatrix} A^{-1}a \\ \mathrm{vec}\left(-\tfrac{1}{2}A^{-1}\right) \end{bmatrix} \cdot \begin{bmatrix} x \\ \mathrm{vec}(xx^T) \end{bmatrix} - \tfrac{1}{2}\left(D\log(2\pi) + \log(|A|) + a^TA^{-1}a\right)\right) \cdot$$

$$\exp\left(\begin{bmatrix} B^{-1}b \\ \mathrm{vec}\left(-\tfrac{1}{2}B^{-1}\right) \end{bmatrix} \cdot \begin{bmatrix} x \\ \mathrm{vec}(xx^T) \end{bmatrix} - \tfrac{1}{2}\left(D\log(2\pi) + \log(|B|) + b^TB^{-1}b\right)\right)$$

$$= \exp\left(\begin{bmatrix} A^{-1}a + B^{-1}b \\ \mathrm{vec}\left(-\tfrac{1}{2}A^{-1}\right) + \mathrm{vec}\left(-\tfrac{1}{2}B^{-1}\right) \end{bmatrix} \begin{bmatrix} x \\ \mathrm{vec}(xx^T) \end{bmatrix} - \tfrac{1}{2}\left(2D\log(2\pi) + \log(|A|) + \log(|B|) + a^TA^{-1}a + b^TB^{-1}b\right)\right)$$

$$= \exp\left(\begin{bmatrix} A^{-1}a + B^{-1}b \\ \mathrm{vec}\left(-\tfrac{1}{2}(A^{-1}+B^{-1})\right) \end{bmatrix} \begin{bmatrix} x \\ \mathrm{vec}(xx^T) \end{bmatrix} - \tfrac{1}{2}\left(2D\log(2\pi) + \log(|A|) + \log(|B|) + a^TA^{-1}a + b^TB^{-1}b\right)\right)$$

↳ note that this is not a valid probability distribution because it is not normalized.

If we now write a distribution

$$\propto \exp\left(\begin{bmatrix} C^{-1}c \\ \mathrm{vec}\left(-\tfrac{1}{2}C^{-1}\right) \end{bmatrix} \begin{bmatrix} x \\ \mathrm{vec}(xx^T) \end{bmatrix} - \tfrac{1}{2}\left(D\log(2\pi) + \log(|C|) + c^TC^{-1}c\right)\right)$$

we see that

$$C^{-1}c = A^{-1}a + B^{-1}b \rightarrow c = C(A^{-1}a + B^{-1}b)$$

and

$$A^{-1} + B^{-1} = C^{-1} \rightarrow C = (A^{-1} + B^{-1})^{-1}$$

The z constant at the front is a scaling constant such that

$$\mathcal{N}(x \mid a, A) \cdot \mathcal{N}(x \mid b, B) = z\,\mathcal{N}(x \mid c, C)$$

We can solve for this by putting z in the exponent and solving:

$$-\frac{1}{2}\left(2\log(z) + D\log(2\pi) + \log(|C|) + c^T C^{-1} c\right) =$$
$$-\frac{1}{2}\left(2D\log(2\pi) + \log(|A|) + \log(|B|) + a^T A^{-1} a + b^T B^{-1} b\right)$$

$$2\log(z) = D\log(2\pi) + \log(|A|) + \log(|B|) + a^T A^{-1} a + b^T B^{-1} b - \log(|C|) - c^T C^{-1} c$$

$$\log(z) = \frac{D}{2}\log(2\pi) + \frac{1}{2}\log(|A|) + \frac{1}{2}\log(|B|) + \frac{1}{2} a^T A^{-1} a + \frac{1}{2} b^T B^{-1} b - \frac{1}{2}\log(|C|) - \frac{1}{2} c^T C^{-1} c$$

$$\log(z) = \log(2\pi)^{\frac{D}{2}} + \log\left(\frac{|A||B|}{|C|}\right)^{\frac{1}{2}} + \frac{1}{2}\left(a^T A^{-1} a + b^T B^{-1} b - c^T C^{-1} c\right)$$

derivations in part a.

$$= \log(2\pi)^{\frac{D}{2}} + \log(|A+B|)^{\frac{1}{2}} + \frac{1}{2}\left[(a-b)^T (A+B)^{-1} (a-b)\right]$$

$$z = (2\pi)^{\frac{D}{2}} + |A+B|^{\frac{1}{2}} + \exp\left(\frac{1}{2}(a-b)^T (A+B)^{-1} (a-b)\right)$$

$$c = \frac{1}{z} = 2\pi^{-\frac{D}{2}} + |A+B|^{-\frac{1}{2}} + \exp\left(-\frac{1}{2}(a-b)^T (A+B)^{-1} (a-b)\right)$$

6.11

$$\mathbb{E}_x[x] = \int x p(x)\, dx$$

$$\mathbb{E}_x[x|y] = \int x p(x|y)\, dx$$

$$\mathbb{E}_y\left[\mathbb{E}_x[x|y]\right] = \int \left(\int x p(x|y)\, dx\right) p(y)\, dy$$

$$= \iint x p(x|y) p(y)\, dx\, dy = \iint x p(x,y)\, dx\, dy$$

move outside because it is a constant in this integral since it is not being integrated over.

$$= \iint x p(x,y)\, dy\, dx = \int x \left(\int p(x,y)\, dy\right) dx =$$

$$= \int x p(x)\, dx = \mathbb{E}_x[x]$$

since $\int p(x,y)\, dy = p(x)$ marginalizing y

6.12

$$y = Ax + b + w \qquad x \sim \mathcal{N}(x|\mu_x, \Sigma_x) \qquad w \sim \mathcal{N}(w|0, Q)$$

a) $p(y|x) = \mathcal{N}(Ax+b, Q)$

$$\mathbb{E}[y|x] = Ax + b + \mathbb{E}[w] = Ax + b + 0 = Ax + b$$

given

$$\mathbb{V}[y|x] = \mathbb{V}[w] = Q$$

b) $p(y) = \mathcal{N}(A\mu_x + b, A\Sigma_x A^T + Q)$ $\qquad \mu_y = A\mu_x + b$

$\Sigma_y = A\Sigma_x A^T + Q$

$$\mathbb{E}[y] = \mathbb{E}[Ax] + b + \mathbb{E}[w] = A\mu_x + b + 0 = A\mu_x + b$$

$$\mathbb{V}[y] = \mathbb{V}[Ax] + \mathbb{V}[w] = A\Sigma_x A^T + Q$$

c)

$$z = Cy + v$$

$$\mathbb{E}[z|y] = Cy + \mathbb{E}[v] = Cy + 0 = Cy$$

$$\mathbb{V}[z|y] = \mathbb{V}[v] = R$$

$$p(z|y) = \mathcal{N}(Cy, R)$$

$$\mathbb{E}[z] = \mathbb{E}[Cy] + \mathbb{E}[v] = C\mathbb{E}[y] + 0 = C(A\mu_x + b)$$

$$\mathbb{V}[z] = \mathbb{V}[Cy] + \mathbb{V}[v] = C\,\mathbb{V}[y]\,C^T + R =$$

$$= C(\Sigma_y)C^T + R = C(A\Sigma_x A^T + Q)C^T + R$$

$$p(z) = \mathcal{N}\left(C(A\mu_x + b),\ C(A\Sigma_x A^T + Q)C^T + R\right)$$

d)

$$p(x|\hat{y}) = \frac{\overbrace{p(\hat{y}|x)}^{p(y|x)}\ p(x)}{\underbrace{p(\hat{y})}_{\text{this is just } p(y)}} \qquad p(x,y) = p(y|x)p(x)$$

We know $p(x,y)$ is Gaussian:

$$p(x,y) = \mathcal{N}\left(\begin{bmatrix}\mu_x \\ \mu_y\end{bmatrix}, \begin{bmatrix}\Sigma_{xx} & \Sigma_{xy} \\ \Sigma_{yx} & \Sigma_{yy}\end{bmatrix}\right)$$

We have $\mu_y = A\mu_x + b$, $\Sigma_{xx} = \mathbb{V}_x[x]$

$$\Sigma_{yy} = \mathbb{V}[y] = \mathbb{V}_{xw}[Ax + \underbrace{b}_{\text{this is a constant so can remove}} + w] = \mathbb{V}_{xw}[Ax + w] = \mathbb{V}_x[Ax] + \mathbb{V}_w[w] + \mathrm{cov}[x,w] + \mathrm{cov}[w,x]$$

$$= A\Sigma_x A^T + Q + \mathrm{cov}[x,w] + \mathrm{cov}[w,x] = A\Sigma_x A^T + Q$$

Since w is independent of x, then $\mathrm{cov}[x,w] = \mathrm{cov}[w,x] = 0$

So far we have

$$p(x,y) = \mathcal{N}\left(\begin{bmatrix}\mu_x \\ A\mu_x + b\end{bmatrix}, \begin{bmatrix}\Sigma_x & \Sigma_{xy} \\ \Sigma_{yx} & A\Sigma_x A^T + Q\end{bmatrix}\right)$$

We now need to find Σ_{xy} and Σ_{yx}

$$\Sigma_{xy} = \mathrm{cov}(x,y) = E[xy^T] - E[x]E[y]^T$$

$$= E[x(Ax+b+w)^T] - E[x]E[Ax+b+w]^T$$

$$= E[xw^T + xb^T + xx^TA^T] - E[x]E[Ax+b+w]$$

$$= E[xw^T] + E[x]b^T + E[xx^T]A^T - \underbrace{E[x]}_{\mu_x}\left(E[Ax]^T + \overset{b^T}{E[b]^T} + \overset{0^T \text{ since } w \sim \mathcal{N}(0,Q)}{E[w]^T}\right)$$

$$E_{xw}[xw^T] = \iint x w^T p(x) p(w) dx dw =$$

$$\int x \left(\int w^T p(w) dw\right) p(x) dx = \int x 0^T dx = 0 \quad \text{(matrix)}$$

$$= \cancel{\mu_x b^T} + E[xx^T]A^T - \mu_x \mu_x^T A^T - \cancel{\mu_x b^T}$$

$$= \left(E[xx^T] - \mu_x \mu_x^T\right) A^T = \Sigma_x A^T$$

$$\Sigma_{yx} = \Sigma_{xy}^T = A\Sigma_x^T$$

so we get

$$p(x,y) = \mathcal{N}\left(\begin{bmatrix}\mu_x \\ A\mu_x + b\end{bmatrix}, \begin{bmatrix}\Sigma_x & \Sigma_x A^T \\ A\Sigma_x & A\Sigma_x A^T + Q\end{bmatrix}\right)$$

Now applying conditioning rules for the Gaussian distribution we get:

$$\mu_{x|y} = \mu_x + \left(\Sigma_x A^T (A\Sigma_x A^T + Q)^{-1} (y - (Ax+b)\right)$$

$$\Sigma_{x|y} = \Sigma_x - \Sigma_x A^T (A\Sigma_x A^T + Q)^{-1} A\Sigma_x$$

6.13

$y = F_X(x) = P(X \le x) = \int_a^x p(t)\,dt$, a is a constant, x is a variable

$0 \le y \le 1$ because a cdf maps x to between 0 and 1.

$$P(Y \le y) = P(F_X(x) \le y) = P(x \le F_X^{-1}(y)) = F_X(F_X^{-1}(y)) = y$$

$F_Y(y) = y$ which is the cdf of the uniform distribution.

also $\frac{d}{dy} F_Y(y) = p(y) = \frac{d}{dy} y = 1$

Anothe way to prove it, have a look in the future:

note this is a function of x not p(x)

We have

$y = F(x)$, so $0 \le y \le 1$ since cdf goes between 0 and 1

We also have

$p(y|x) = \delta(y - F(x))$ where δ is the Dirac delta function.

Intuition for other proof although it doesn't work because $p(y|x) \neq 1$ but $p(y|x) = \delta(y - f(x))$

this is the Dirac delta function and not 1

$$p(y) = \int p(y,x)\,dx = \int p(y|x)\,p(x)\,dx = \int p(x)\,dx = 1$$

this is wrong.

We see that $p(y)$ is uniform and that

$$\int_0^1 p(y)\,dy = \int_0^1 1\,dy = y\Big]_0^1 = 1$$

, so it is a valid distribution.